



EAST TEXAS A&M
— UNIVERSITY —

CSCI 352.61E

DIGITAL FORENSICS

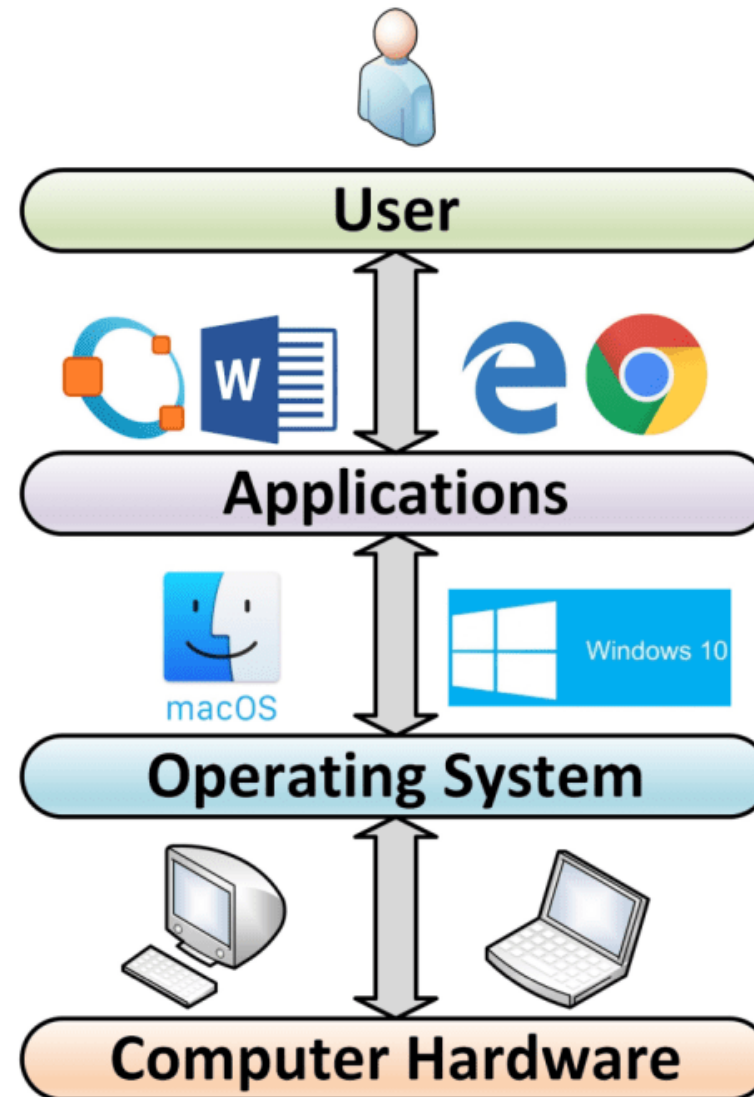
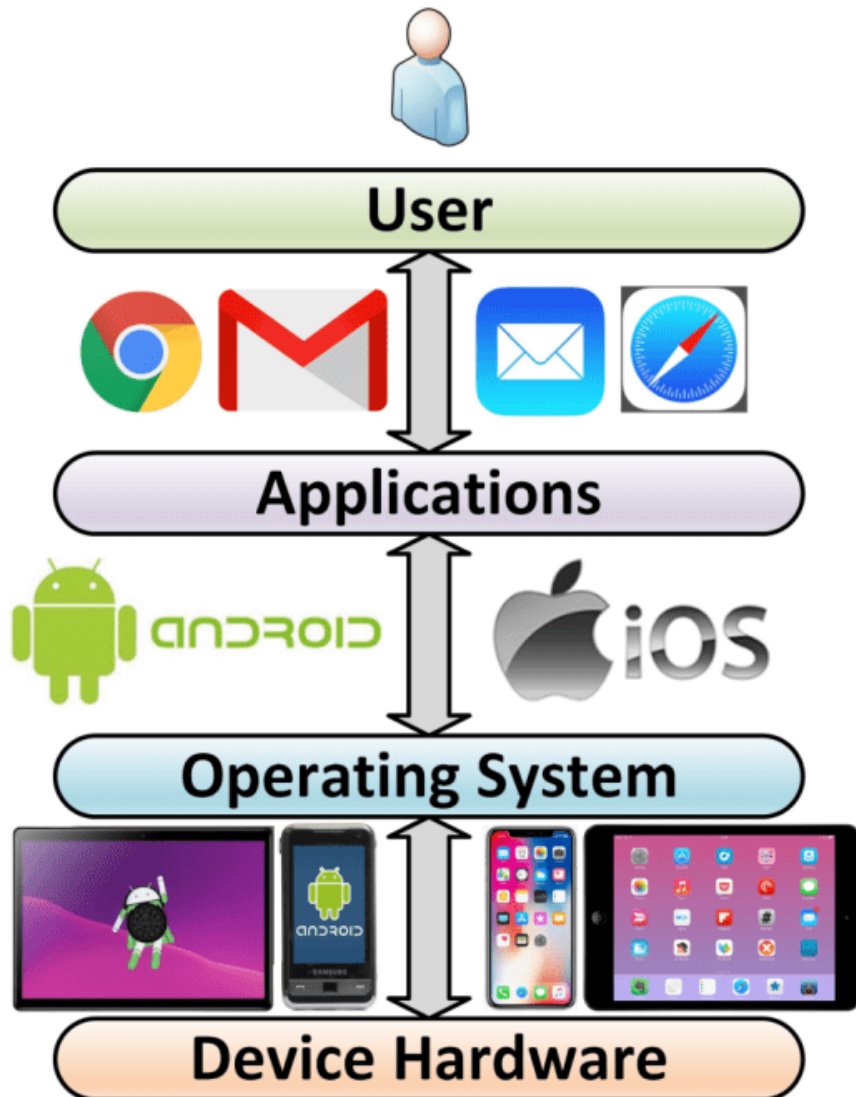
INVESTIGATE DATA LEAKAGE CASE

Keywords: Windows Security Event Logs, XML

GOALS

- Understand evidence generated by OS
- Understand XML format
- Know how to extract information from XML files

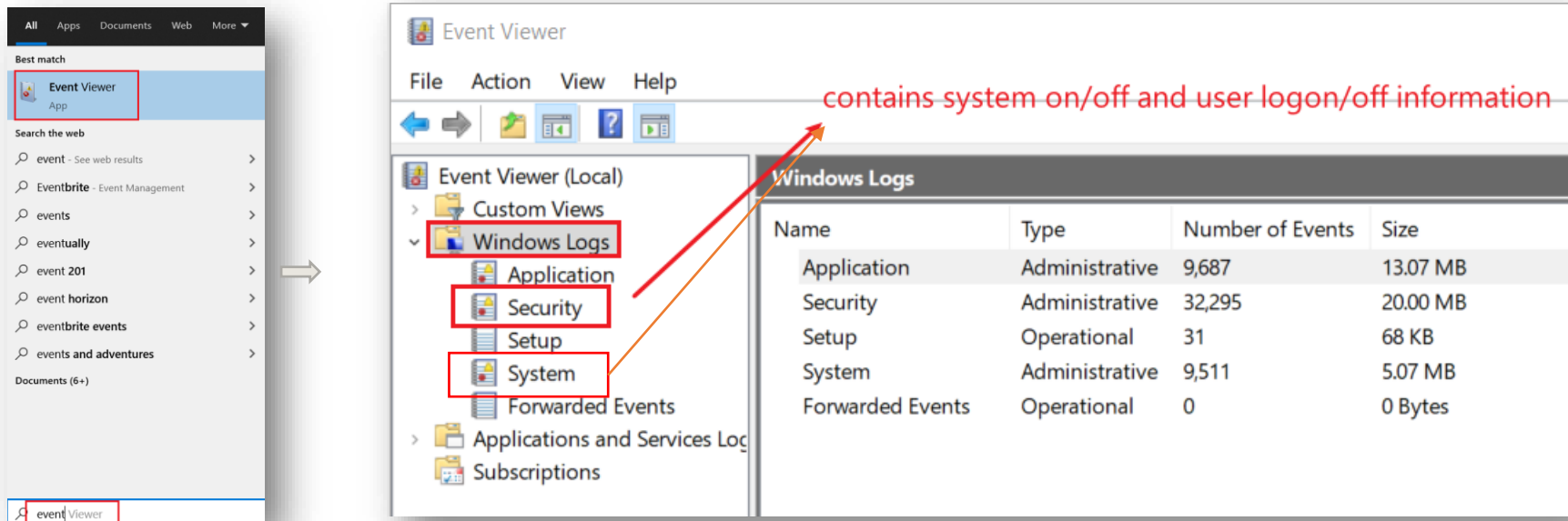
LAYERS OF A COMPUTER SYSTEM



- Evidence is generated from each layer
- User activities= Applications+ OS+ Disk/Memory

12. LIST ALL TRACES ABOUT THE SYSTEM ON/OFF AND THE USER LOGON/LOGOFF (IT SHOULD BE CONSIDERED ONLY DURING A TIME RANGE BETWEEN 09:00 AND 18:00 IN THE TIMEZONE (EASTERN TIME) FROM QUESTION 4.

- The Windows event log is a detailed record of system, security and application notifications stored by the Windows
- It's a useful tool for
 - troubleshooting all kinds of different Windows problems
 - forensics analysis



Event Viewer

FileActionViewHelp

Event Viewer (Local)

Custom Views

Windows Logs

Application

Security

Setup

System

Forwarded Events

Applications and Services Logs

Subscriptions

Security

Number of events: 32,345 (!) New events available

Keywords	Date and Time	Source	Event ID	Task Category
Audit Success	1/26/2021 8:54:03 AM	Microsoft Windows security auditing.	4672	Special Logon
Audit Success	1/26/2021 8:54:03 AM	Microsoft Windows security auditing.	4624	Logon
Audit Success	1/26/2021 8:50:11 AM	Microsoft Windows security auditing.	4672	Special Logon
Audit Success	1/26/2021 8:50:11 AM	Microsoft Windows security auditing.	4624	Logon
Audit Success	1/26/2021 8:45:07 AM	Microsoft Windows security auditing.	4798	User Account Management
Audit Success	1/26/2021 8:45:07 AM	Microsoft Windows security auditing.	4798	User Account Management

Event 4624, Microsoft Windows security auditing.

General

Details

An account was successfully logged on.

Subject:

Security ID:SYSTEM

Account Name:RYZEN3790-XU\$

Account Domain:WORKGROUP

Logon ID:0x3E7

Logon Information:

Logon Type:5

Restricted Admin Mode:-

Virtual Account:No

Elevated Token:Yes

Impersonation Level:Impersonation

Log Name:Security

Source:Microsoft Windows security ;

Event ID:4624

Level:Information

User:N/A

OpCode:Info

More Information:[Event Log Online Help](#)

Logged:1/26/2021 8:54:03 AM

Task Category:Logon

Keywords:Audit Success

Computer:ryzen3790-xu

Logon event 4624

HOW TO SEARCH EVENTS?

<https://www.ultimatewindowssecurity.com/securitylog/encyclopedia/default.aspx>

ultimatewindowssecurity.com/securitylog/encyclopedia/default.aspx

SharePoint SQL Server Exchange → **LOGbinder** → YOUR SIEM

SUPERCHARGE
for Windows Event Collection

Home > Security Log > Encyclopedia

 **Randy Franklin Smith's**
ULTIMATE IT SECURITY .COM

February 2021 Patch Monday "Patch Monday" **LOGbinder** User name: Password: Re

Follow @randyfrsmith 3,615 followers

Security Log Windows SharePoint SQL Server Exchange Training Tools Webinars Blog Forum

Webinars Training Encyclopedia Quick Reference Book

Encyclopedia

- All Event IDs
- Audit Policy

Go To Event ID: Go

Security Log Quick Reference Chart

Windows Security Log Events

☐ All Sources
☒ **Windows Audit**

☐ SharePoint Audit (LOGbinder for SharePoint)
☐ SQL Server Audit (LOGbinder for SQL Server)
☐ Exchange Audit (LOGbinder for Exchange)
☐ Sysmon (MS Sysinternals Sysmon)

Windows Audit Categories:
System
Subcategories:
All subcategories

Windows Versions:
☐ All events
☐ Win2000, XP and Win2003 only
☒ Win2008, Win2012R2, Win2016 and Win10+, Win2019

Category: System

Windows 4608 Windows is starting up
Windows 4609 Windows is shutting down
Windows 4610 An authentication package has been loaded by the Local Security Authority
Windows 4611 A trusted logon process has been registered with the Local Security Authority

OVERVIEW APPROACH

- Get a copy of security event log
- Analyze the event log
 - Install XML parser (xmlstarlet)
 - Validating Security Event Log XML documents
 - Display element structure of XML document
 - Format XML document
 - Query XML documents
 - Modify or edit XML documents

GET A COPY OF SECURITY EVENT LOG

Exam security event logs for logon/logoff

```
root@kali:~/lab#  
root@kali:~/lab# evtx_dump.py Security.evtx > SecurityEvt.xml  
root@kali:~/lab#  
root@kali:~/lab# ls -l SecurityEvt.xml  
-rw-r--r-- 1 root root 1555142 Nov 21 19:12 SecurityEvt.xml  
root@kali:~/lab#
```

Notes: If you are not in root, you will see an error message. You need to check the permission of the lab folder
See next slide debugs error messages if you see any.

switch to root if there is a permission problem

```
(student@kali80)-[~/lab]
$ sudo evt_x_dump.py Security.evtx > SecurityEvt.xml
zsh: permission denied: SecurityEvt.xml

(student@kali80)-[~/lab]
$ sudo su
(root@kali80)-[/home/student/lab]
# evt_x_dump.py Security.evtx > SecurityEvt.xml

(root@kali80)-[/home/student/lab]
# ls Security.*
Security.evtx

(root@kali80)-[/home/student/lab]
# ls Security*.*
Security.evtx SecurityEvt.xml
```

WHAT IS XML

- An XML file is an eXtensible Markup Language file that structures data for storage and transport.
- XML files can be used for various purposes
 - Data transfer: XML files can store and retrieve data for different applications.
 - Formatting documents: XML files can define how web pages or other documents are displayed.
 - Web searching: XML files can help search engines index and rank web pages
 - Creating layouts: XML files can design the user interface of mobile applications.
 - Storing configuration data: XML files can store the settings and preferences of an application.

INSTALL XML PARSER

XMLSTARLET

- Is a set of command line utilities (tools)
 - transform, query, validate, and edit **XML** documents and files
- Using simple set of shell commands
 - Similar to *grep, sed, awk*
- <http://xmlstar.sourceforge.net/doc>

Install *xmlstarlet*



```
root@kali:~/lab#  
root@kali:~/lab# apt-get install xmlstarlet  
Reading package lists... Done  
Building dependency tree  
Reading state information... Done  
xmlstarlet is already the newest version (1.6.1-2.1).
```

```
root@kali:~/lab#
```

```
root@kali:~/lab# xmlstarlet --help
```

XMLStarlet Toolkit: Command line utilities for XML

Usage: xmlstarlet [<options>] <command> [<cmd-options>]

where <command> is one of:

- | | | |
|-------|----------------|--|
| ed | (or edit) | - Edit/Update XML document(s) |
| sel | (or select) | - Select data or query XML document(s) (XPath, etc) ← |
| tr | (or transform) | - Transform XML document(s) using XSLT |
| val | (or validate) | - Validate XML document(s) (well-formed/DTD/XSD/RelaxNG) |
| fo | (or format) | - Format XML document(s) |
| el | (or elements) | - Display element structure of XML document ← |
| c14n | (or canonic) | - XML canonicalization |
| ls | (or list) | - List directory as XML |
| esc | (or escape) | - Escape special XML characters |
| unesc | (or unescape) | - Unescape special XML characters |
| pyx | (or xmln) | - Convert XML into PYX format (based on ESIS - ISO 8879) |
| p2x | (or depyx) | - Convert PYX into XML |

<options> are:

- | | |
|--------------------|---|
| -q or --quiet | - no error output |
| --doc-namespace | - extract namespace bindings from input doc (default) |
| --no-doc-namespace | - don't extract namespace bindings from input doc |
| --version | - show version |
| --help | - show help |

Wherever file name mentioned in command help it is assumed that URL can be used instead as well.

Type: xmlstarlet <command> --help <ENTER> for command help

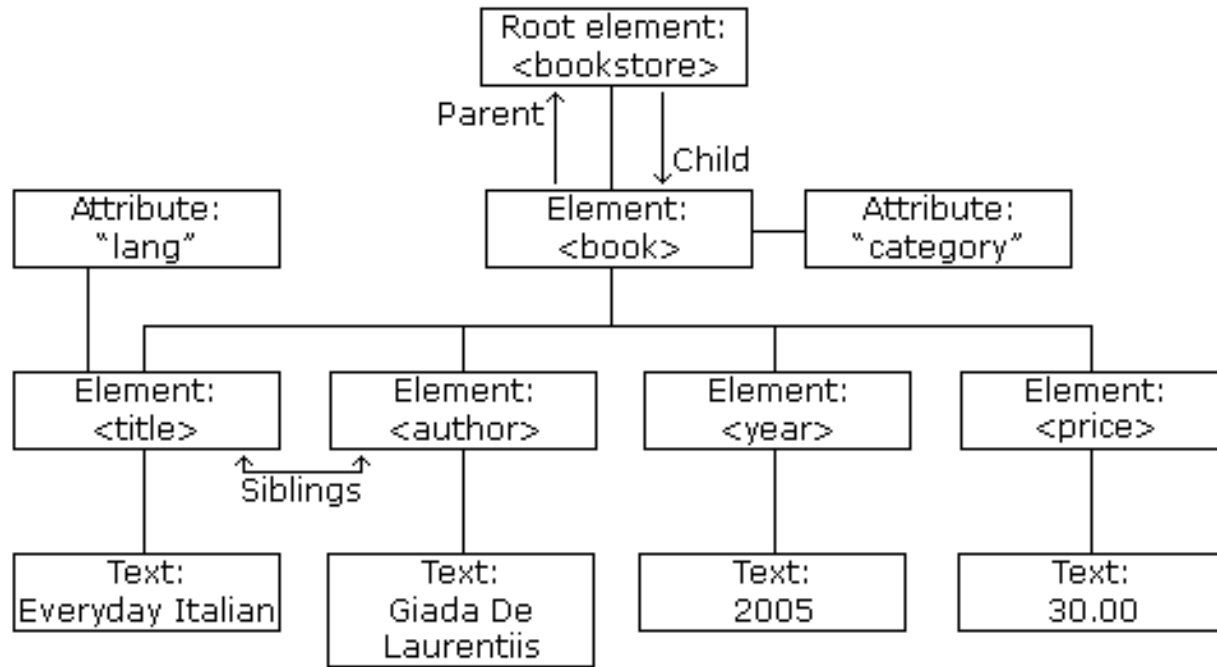
XMLStarlet is a command line toolkit to query/edit/check/transform XML documents (for more information see <http://xmlstar.sourceforge.net/>)

```
root@kali:~/lab#
```

UTF-8 is capable of encoding all 1,112,064 characters using one to four one-byte (8-bit) code units.

An Example XML Document

The image above represents books in this XML:



```
<?xml version="1.0" encoding="UTF-8"?>
<bookstore>
  <book category="cooking">
    <title lang="en">Everyday Italian</title>
    <author>Giada De Laurentiis</author>
    <year>2005</year>
    <price>30.00</price>
  </book>
  <book category="children">
    <title lang="en">Harry Potter</title>
    <author>J K. Rowling</author>
    <year>2005</year>
    <price>29.99</price>
  </book>
  <book category="web">
    <title lang="en">Learning XML</title>
    <author>Erik T. Ray</author>
    <year>2003</year>
    <price>39.95</price>
  </book>
</bookstore>
```

Show *SecurityEvt.xml*

```
root@kali:~/lab# cat SecurityEvt.xml | head -n 25
<?xml version="1.1" encoding="utf-8" standalone="yes" ?>

<Events>
  <Event xmlns="http://schemas.microsoft.com/win/2004/08/events/event"><System><Provi
    der Name="Microsoft-Windows-Security-Auditing" Guid="{54849625-5478-4994-a5ba-3e3b0
    328c30d}"></Provider>
    <EventID Qualifiers="">4608</EventID>
    <Version>0</Version>
    <Level>0</Level>
    <Task>12288</Task>
    <Opcode>0</Opcode>
    <Keywords>0x8020000000000000</Keywords>
    <TimeCreated SystemTime="2015-03-25 10:15:35.248869"></TimeCreated>
    <EventRecordID>1</EventRecordID>
    <Correlation ActivityID="" RelatedActivityID=""></Correlation>
    <Execution ProcessID="464" ThreadID="468"></Execution>
    <Channel>Security</Channel>
    <Computer>37L4247F27-25</Computer>
    <Security UserID=""></Security>
    </System>
    <EventData></EventData>
  </Event>

  <Event xmlns="http://schemas.microsoft.com/win/2004/08/events/event"><System><Provi
    der Name="Microsoft-Windows-Security-Auditing" Guid="{54849625-5478-4994-a5ba-3e3b0
    328c30d}"></Provider>
    <EventID Qualifiers="">4624</EventID>
    <Version>0</Version>
    <Level>0</Level>
  root@kali:~/lab#
```

which **table**?

```
<table>
  <tr>
    <td>Apples</td>
    <td>Bananas</td>
  </tr>
</table>
```

```
<table>
  <name>African Coffee
Table</name>
  <width>80</width>
  <length>120</length>
</table>
```

- **XML Namespaces** provide a method to avoid element name conflicts.
 - defined by an **xmlns** attribute in the start tag of an element.
 - **xmlns:prefix="URI"**.

VALIDATING XML DOCUMENTS

```
root@kali:~/lab#  
root@kali:~/lab# xmlstarlet val SecurityEvt.xml  
SecurityEvt.xml - valid  
root@kali:~/lab#
```

DISPLAY **E**LEMENT STRUCTURE OF XML DOCUMENT

or elements

Show the element
structure of
SecurityEvt.xml

```
root@kali:~/lab#  
root@kali:~/lab# xmlstarlet el -u SecurityEvt.xml  
SecurityEvt.xml:1.20: Unsupported version '1.1'  
<?xml version="1.1" encoding="utf-8" standalone="yes" ?>  
^  
Events  
Events/Event  
Events/Event/EventData  
Events/Event/EventData/Data  
Events/Event/System  
Events/Event/System/Channel  
Events/Event/System/Computer  
Events/Event/System/Correlation  
Events/Event/System/EventID  
Events/Event/System/EventRecordID  
Events/Event/System/Execution  
Events/Event/System/Keywords  
Events/Event/System/Level  
Events/Event/System/Opcode  
Events/Event/System/Provider  
Events/Event/System/Security  
Events/Event/System/Task  
Events/Event/System/TimeCreated  
Events/Event/System/Version  
Events/Event/UserData  
Events/Event/UserData/ServiceShutdown  
root@kali:~/lab#
```

- **el**: Display element structure of XML document
- **-u**: print out sorted unique lines

XML structure



/bin/bash 112x31

root@kali:~/lab#

root@kali:~/lab# `xmlstarlet el --help`

XMLStarlet Toolkit: Display element structure of XML document

Usage: `xmlstarlet el [<options>] <xml-file>`

where

`<xml-file>` - input XML document file name (stdin is used if missing)

`<options>` is one of:

-a - show attributes as well

-v - show attributes and their values

-u - print out sorted unique lines ←

-d<n> - print out sorted unique lines up to depth <n>

XMLStarlet is a command line toolkit to query/edit/check/transform

XML documents (for more information see <http://xmlstar.sourceforge.net/>)

root@kali:~/lab#

Show the element (attributes) structure of *SecurityEvt.xml*

```
root@kali:~/lab#  
root@kali:~/lab# xmlstarlet el -a SecurityEvt.xml | head  
SecurityEvt.xml:1.20: Unsupported version '1.1'  
<?xml version="1.1" encoding="utf-8" standalone="yes" ?>  
^  
Events  
Events/Event  
Events/Event/@xmlns ← -a: show attributes  
Events/Event/System  
Events/Event/System/Provider  
Events/Event/System/Provider/@Name ←  
Events/Event/System/Provider/@Guid ←  
Events/Event/System/EventID  
Events/Event/System/EventID/@Qualifiers ←  
Events/Event/System/Version  
root@kali:~/lab#
```

`<person gender="female">` ← Attribute
`<firstname>Anna</firstname>`
`<lastname>Smith</lastname>` ← Element
`</person>`

Show element attributes and their values

```
root@kali:~/lab#  
root@kali:~/lab# xmlstarlet el -v SecurityEvt.xml 2>/dev/null | head  
Events  
Events/Event[@xmlns='http://schemas.microsoft.com/win/2004/08/events/event']  
Events/Event/System  
Events/Event/System/Provider[@Name='Microsoft-Windows-Security-Auditing' and  
@Guid='{54849625-5478-4994-a5ba-3e3b0328c30d}']  
Events/Event/System/EventID[@Qualifiers='']  
Events/Event/System/Version  
Events/Event/System/Level  
Events/Event/System/Task  
Events/Event/System/Opcode  
Events/Event/System/Keywords  
root@kali:~/lab#
```

`-v` : show attributes and their values

FORMAT XML DOCUMENT

```
root@kali:~/lab#
root@kali:~/lab# xmlstarlet fo --help
XMLStarlet Toolkit: Format XML document
Usage: xmlstarlet fo [<options>] <xml-file>
where <options> are
  -n or --noindent           - do not indent
  -t or --indent-tab         - indent output with tabulation
  -s or --indent-spaces <num> - indent output with <num> spaces
  -o or --omit-decl         - omit xml declaration <?xml version="1.0"?>
  --net                      - allow network access
  -R or --recover           - try to recover what is parsable
  -D or --dropdtd           - remove the DOCTYPE of the input docs
  -C or --nocdata           - replace cdata section with text nodes
  -N or --nsclean           - remove redundant namespace declarations
  -e or --encode <encoding> - output in the given encoding (utf-8, unicode...)
  -H or --html              - input is HTML
  -h or --help              - print help
```

```
root@kali:~/lab#  
root@kali:~/lab# xmlstarlet fo SecurityEvt.xml | head -n 15  
SecurityEvt.xml:1.20: Unsupported version '1.1'  
<?xml version="1.1" encoding="utf-8" standalone="yes" ?>  
^  
<?xml version="1.1" encoding="utf-8" standalone="yes"?>  
<Events>  
  <Event xmlns="http://schemas.microsoft.com/win/2004/08/events/event">  
    <System>  
      <Provider Name="Microsoft-Windows-Security-Auditing" Guid="{5484962  
        <EventID Qualifiers="">4608</EventID> ←  
        <Version>0</Version>  
        <Level>0</Level>  
        <Task>12288</Task>  
        <Opcode>0</Opcode>  
        <Keywords>0x8020000000000000</Keywords>  
        <TimeCreated SystemTime="2015-03-25 10:15:35.248869"/> ←  
        <EventRecordID>1</EventRecordID>  
        <Correlation ActivityID="" RelatedActivityID=""/>  
        <Execution ProcessID="464" ThreadID="468"/>  
root@kali:~/lab#
```

**SELECT DATA OR QUERY XML
DOCUMENT(S) (XPATH, ETC)**

```
root@kali:~/lab#
```

```
root@kali:~/lab# xmlstarlet sel --help
```

```
XMLStarlet Toolkit: Select from XML document(s)
```

```
Usage: xmlstarlet sel <global-options> {<template>} [ <xml-file> ... ] ←
```

```
where
```

```
<global-options> - global options for selecting
```

```
<xml-file> - input XML document file name/uri (stdin is used if missing)
```

```
<template> - template for querying XML document with following syntax:
```

```
<global-options> are:
```

- | | |
|---------------------------|--|
| -Q or --quiet | - do not write anything to standard output. |
| -C or --comp | - display generated XSLT |
| -R or --root | - print root element <xsl-select> |
| -T or --text | - output is text (default is XML) |
| -I or --indent | - indent output |
| -D or --xml-decl | - do not omit xml declaration line |
| -B or --noblanks | - remove insignificant spaces from XML tree |
| -E or --encode <encoding> | - output in the given encoding (utf-8, unicode...) |
| -N <name>=<value> | - predefine namespaces (name without 'xmlns:')
ex: xsql=urn:oracle-xsql
Multiple -N options are allowed. |
| --net | - allow fetch DTDs or entities over network |
| --help | - display help |

Syntax for templates: `-t|--template <options>`

where <options>

What is XPATH expression?

- c or --copy-of <xpath> - print copy of XPATH expression
- v or --value-of <xpath> - print value of XPATH expression
- o or --output <string> - output string literal
- n or --nl - print new line
- f or --inp-name - print input file name (or URL)
- m or --match <xpath> - match XPATH expression
- var <name> <value> --break or
- var <name>=<value> - declare a variable (referenced by \$name)
- i or --if <test-xpath> - check condition <xsl:if test="test-xpath">
- elif <test-xpath> - check condition if previous conditions failed
- else - check if previous conditions failed
- e or --elem <name> - print out element <xsl:element name="name">
- a or --attr <name> - add attribute <xsl:attribute name="name">
- b or --break - break nesting
- s or --sort op xpath - sort in order (used after -m) where
op is X:Y:Z,
 - X is A - for order="ascending"
 - X is D - for order="descending"
 - Y is N - for data-type="numeric"
 - Y is T - for data-type="text"
 - Z is U - for case-order="upper-first"
 - Z is L - for case-order="lower-first"

XPATH EXPRESSIONS

- *Xpath* uses path expressions to select nodes or node-sets in an XML document.
- *Xpath* expressions look very much like the expressions you see when you work with a traditional computer file system.
- **Must** provide XML namespace for queuing data

```
/bin/bash 116x23
root@kali:~/lab#
root@kali:~/lab# xmlstarlet fo SecurityEvt.xml | head -n 15
SecurityEvt.xml:1.20: Unsupported version '1.1'
<?xml version="1.1" encoding="utf-8" standalone="yes" ?>
^
<?xml version="1.1" encoding="utf-8" standalone="yes"?>
<Events>
  <Event xmlns="http://schemas.microsoft.com/win/2004/08/events/event">
    <System>
      <Provider Name="Microsoft-Windows-Security-Auditing" Guid="{5484962
      <EventID Qualifiers="">4608</EventID> ←
      <Version>0</Version>
      <Level>0</Level>
      <Task>12288</Task>
      <Opcode>0</Opcode>
      <Keywords>0x8020000000000000</Keywords>
      <TimeCreated SystemTime="2015-03-25 10:15:35.248869"/> ←
      <EventRecordID>1</EventRecordID>
      <Correlation ActivityID="" RelatedActivityID=""/>
      <Execution ProcessID="464" ThreadID="468"/>
root@kali:~/lab#
```

XPATH EXPRESSIONS

- First Event node
 - `//x:Event[1]`
- Last event
 - `//x:Event[last()]`
- First two events
 - `//x:Event[position()<3]`
- *EventID* with an attribute named *Qualifiers*
 - `//x:EventID[@Qualifiers]`

“//” operator, a convenient shorthand for selecting nodes at any level in a document.

```
x: namespace variable, e.g.,
/bin/bash 116x23
root@kali:~/lab#
root@kali:~/lab# xmlstarlet fo SecurityEvt.xml | head -n 15
SecurityEvt.xml:1.20: Unsupported version '1.1'
<?xml version="1.1" encoding="utf-8" standalone="yes" ?>
^
<?xml version="1.1" encoding="utf-8" standalone="yes"?>
<Events>
  <Event xmlns="http://schemas.microsoft.com/win/2004/08/events/event">
    <System>
      <Provider Name="Microsoft-Windows-Security-Auditing" Guid="{5484962
        <EventID Qualifiers="">4608</EventID> ←
        <Version>0</Version>
        <Level>0</Level>
        <Task>12288</Task>
        <Opcode>0</Opcode>
        <Keywords>0x8020000000000000</Keywords>
        <TimeCreated SystemTime="2015-03-25 10:15:35.248869"/> ←
        <EventRecordID>1</EventRecordID>
        <Correlation ActivityID="" RelatedActivityID=""/>
        <Execution ProcessID="464" ThreadID="468"/>
root@kali:~/lab#
```

Select the first event by using template to match the xpath and print the copy of the context node

```
root@kali:~/lab#  
root@kali:~/lab# xmlstarlet sel -N x="http://schemas.microsoft.com/win/2004/08/events/event"  
-t -m "//x:Event[1]" -c . SecurityEvt.xml  
SecurityEvt.xml:1.20: Unsupported version '1.1'  
<?xml version="1.1" encoding="utf-8" standalone="yes" ?>  
^  
<Event xmlns="http://schemas.microsoft.com/win/2004/08/events/event"><System><Provider Name="Microsoft-Windows-Security-Auditing" Guid="{54849625-5478-4994-a5ba-3e3b0328c30d}"/>  
<EventID Qualifiers="">4608</EventID>  
<Version>0</Version>  
<Level>0</Level>  
<Task>12288</Task>  
<Opcode>0</Opcode>  
<Keywords>0x8020000000000000</Keywords>  
<TimeCreated SystemTime="2015-03-25 10:15:35.248869"/>  
<EventRecordID>1</EventRecordID>  
<Correlation ActivityID="" RelatedActivityID=""/>  
<Execution ProcessID="464" ThreadID="468"/>  
<Channel>Security</Channel>  
<Computer>37L4247F27-25</Computer>  
<Security UserID=""/>  
</System>  
<EventData/>  
root@kali:~/lab#
```

-N: namespace
x: namespace variable

- The **dot** (".") : the "context item expression" because it refers to the context item.
 - This could be a node (such as an element, attribute, or text node), or an atomic value (such as a string, number, or boolean).
 - When it's a node, it's also called the context node.
- **-c** or **--copy-of <xpath>**: print copy of XPATH expression
- **options**
 - **-m**: match XPATH expression
 - **-t**: template <options>

Select TimeCreated of events and print the context node

```
root@kali:~/lab#  
root@kali:~/lab# xmlstarlet sel -N x="http://schemas.microsoft.com/win/2004/08/events/event" -t -m "//x:TimeCreated"  
-c . -n SecurityEvt.xml | head  
SecurityEvt.xml:1.20: Unsupported version '1.1'  
<?xml version="1.1" encoding="utf-8" standalone="yes" ?>  
^  
<TimeCreated xmlns="http://schemas.microsoft.com/win/2004/08/events/event" SystemTime="2015-03-25 10:15:35.248869"/>  
<TimeCreated xmlns="http://schemas.microsoft.com/win/2004/08/events/event" SystemTime="2015-03-25 10:15:35.311270"/>  
<TimeCreated xmlns="http://schemas.microsoft.com/win/2004/08/events/event" SystemTime="2015-03-25 10:15:37.089672"/>  
<TimeCreated xmlns="http://schemas.microsoft.com/win/2004/08/events/event" SystemTime="2015-03-25 10:15:37.370474"/>  
<TimeCreated xmlns="http://schemas.microsoft.com/win/2004/08/events/event" SystemTime="2015-03-25 10:15:37.370474"/>  
<TimeCreated xmlns="http://schemas.microsoft.com/win/2004/08/events/event" SystemTime="2015-03-25 10:15:37.370474"/>  
<TimeCreated xmlns="http://schemas.microsoft.com/win/2004/08/events/event" SystemTime="2015-03-25 10:15:37.370474"/>  
<TimeCreated xmlns="http://schemas.microsoft.com/win/2004/08/events/event" SystemTime="2015-03-25 10:15:37.370474"/>  
<TimeCreated xmlns="http://schemas.microsoft.com/win/2004/08/events/event" SystemTime="2015-03-25 10:15:37.370474"/>  
<TimeCreated xmlns="http://schemas.microsoft.com/win/2004/08/events/event" SystemTime="2015-03-25 10:15:37.370474"/>  
root@kali:~/lab#
```

-n: print new line

4608 (Windows is starting up)
1100 (service shutdown)
4624 (successful logon)
4634 (logoff)
4625 (logon failure)
4647 (a user initiated the logoff process)...

Select a property of EventID and print the context node

```
root@kali:~/lab#  
root@kali:~/lab# xmlstarlet sel -N x="http://schemas.microsoft.com/win/2004/08/events/event" -t -m  
"//x:EventID[@Qualifiers]" -c . -n SecurityEvt.xml | head  
SecurityEvt.xml:1.20: Unsupported version '1.1'  
<?xml version="1.1" encoding="utf-8" standalone="yes" ?>  
^  
<EventID xmlns="http://schemas.microsoft.com/win/2004/08/events/event" Qualifiers="">4608</EventID>  
<EventID xmlns="http://schemas.microsoft.com/win/2004/08/events/event" Qualifiers="">4624</EventID>  
<EventID xmlns="http://schemas.microsoft.com/win/2004/08/events/event" Qualifiers="">4902</EventID>  
<EventID xmlns="http://schemas.microsoft.com/win/2004/08/events/event" Qualifiers="">4731</EventID>  
<EventID xmlns="http://schemas.microsoft.com/win/2004/08/events/event" Qualifiers="">4735</EventID>  
<EventID xmlns="http://schemas.microsoft.com/win/2004/08/events/event" Qualifiers="">4731</EventID>  
<EventID xmlns="http://schemas.microsoft.com/win/2004/08/events/event" Qualifiers="">4735</EventID>  
<EventID xmlns="http://schemas.microsoft.com/win/2004/08/events/event" Qualifiers="">4731</EventID>  
<EventID xmlns="http://schemas.microsoft.com/win/2004/08/events/event" Qualifiers="">4735</EventID>  
<EventID xmlns="http://schemas.microsoft.com/win/2004/08/events/event" Qualifiers="">4731</EventID>  
root@kali:~/lab#
```

-n: print new line

SUBSTITUTE *XMLSTARLET* *SEL*
RESULTS USING *SED*

STREAM EDITOR

- Used to perform basic text transformations on an input stream (a file or input from a pipeline).
- Can apply regular expression

Substitute *morning* in text stream with *afternoon*

```
root@kali:~/lab#  
root@kali:~/lab# echo good morning  
good morning  
root@kali:~/lab# echo good morning | sed "s/morning/afternoon/"  
good afternoon  
root@kali:~/lab#
```

substitute "morning" with "afternoon"

Substitute *morning* in text stream with *afternoon*

```
root@kali:~/lab# /bin/bash 100x24
root@kali:~/lab# echo good morning > oldfile.txt
root@kali:~/lab# cat oldfile.txt
good morning
root@kali:~/lab# sed "s/morning/afternoon/" < oldfile.txt >newfile.txt
root@kali:~/lab# cat newfile.txt
good afternoon
root@kali:~/lab#
```

```
root@kali:~/lab#  
root@kali:~/lab# echo good morning morning | sed "s/morning/afternoon/"  
good afternoon morning  
root@kali:~/lab#  
root@kali:~/lab# echo good morning morning | sed "s/morning/afternoon/g"  
good afternoon afternoon
```

/g - Global replacement

4608 (Windows is starting up)

1100 (service shutdown)

4624 (successful logon)

4634 (logoff)

4625 (logon failure)

4647 (a user initiated the logoff process)...

```
/bin/bash 100x24
root@kali:~/lab#
root@kali:~/lab# xmlstarlet sel -N x="http://schemas.microsoft.com/win/2004/08/events/event" -t -m
"//x:EventID[@Qualifiers]" -c . -n SecurityEvt.xml | sed "s/4608/Windows is starting up/" | head
SecurityEvt.xml:1.20: Unsupported version '1.1'
<?xml version="1.1" encoding="utf-8" standalone="yes" ?>
    ^
<EventID xmlns="http://schemas.microsoft.com/win/2004/08/events/event" Qualifiers="">Windows is star
ting up</EventID> ←
<EventID xmlns="http://schemas.microsoft.com/win/2004/08/events/event" Qualifiers="">4624</EventID>
<EventID xmlns="http://schemas.microsoft.com/win/2004/08/events/event" Qualifiers="">4902</EventID>
<EventID xmlns="http://schemas.microsoft.com/win/2004/08/events/event" Qualifiers="">4731</EventID>
<EventID xmlns="http://schemas.microsoft.com/win/2004/08/events/event" Qualifiers="">4735</EventID>
<EventID xmlns="http://schemas.microsoft.com/win/2004/08/events/event" Qualifiers="">4731</EventID>
<EventID xmlns="http://schemas.microsoft.com/win/2004/08/events/event" Qualifiers="">4735</EventID>
<EventID xmlns="http://schemas.microsoft.com/win/2004/08/events/event" Qualifiers="">4731</EventID>
<EventID xmlns="http://schemas.microsoft.com/win/2004/08/events/event" Qualifiers="">4735</EventID>
<EventID xmlns="http://schemas.microsoft.com/win/2004/08/events/event" Qualifiers="">4731</EventID>
root@kali:~/lab#
```

EDIT XML DOCUMENT(S)

```
root@kali:~/lab# xmlstarlet ed --help
```

```
XMLStarlet Toolkit: Edit XML document(s)
```

```
Usage: xmlstarlet ed <global-options> {<action>} [ <xml-file-or-uri> ... ]
```

```
where
```

```
<global-options> - global options for editing
```

```
<xml-file-or-uri> - input XML document file name/uri (stdin otherwise)
```

```
<global-options> are:
```

```
-P, or -S - preserve whitespace nodes.
```

```
(or --pf, --ps) Note that space between attributes is not preserved
```

```
-O (or --omit-decl) - omit XML declaration (<?xml ...?>)
```

```
-L (or --inplace) - edit file inplace
```

```
-N <name>=<value> - predefine namespaces (name without 'xmlns:')  
ex: xsql=urn:oracle-xsql
```

```
Multiple -N options are allowed.
```

```
-N options must be last global options.
```

```
--net - allow network access
```

```
--help or -h - display help
```

```
where <action>
```

```
-d or --delete <xpath>
```

```
--var <name> <xpath>
```

```
-i or --insert <xpath> -t (--type) elem|text|attr -n <name> [-v (--value) <value>]
```

```
-a or --append <xpath> -t (--type) elem|text|attr -n <name> [-v (--value) <value>]
```

```
-s or --subnode <xpath> -t (--type) elem|text|attr -n <name> [-v (--value) <value>]
```

```
-m or --move <xpath1> <xpath2>
```

```
-r or --rename <xpath1> -v <new-name>
```

```
-u or --update <xpath> -v (--value) <value>
```

Show the first event and we will make change the event

```
/bin/bash 98x30
root@kali:~/lab#
root@kali:~/lab# xmlstarlet sel -N x="http://schemas.microsoft.com/win/2004/08/events/event" -t -m
"//x:Event[1]" -c . SecurityEvt.xml 2>/dev/null > SecurityEvent1.xml
root@kali:~/lab#
root@kali:~/lab# cat SecurityEvent1.xml
<Event xmlns="http://schemas.microsoft.com/win/2004/08/events/event"><System><Provider Name="Micro
soft-Windows-Security-Auditing" Guid="{54849625-5478-4994-a5ba-3e3b0328c30d}"/>
<EventID Qualifiers="">4608</EventID>
<Version>0</Version>
<Level>0</Level> ← Let's remove the node
<Task>12288</Task>
<Opcode>0</Opcode>
<Keywords>0x8020000000000000</Keywords>
<TimeCreated SystemTime="2015-03-25 10:15:35.248869"/>
<EventRecordID>1</EventRecordID>
<Correlation ActivityID="" RelatedActivityID=""/>
<Execution ProcessID="464" ThreadID="468"/>
<Channel>Security</Channel>
<Computer>37L4247F27-25</Computer>
<Security UserID=""/>
</System>
<EventData/>
</Event>root@kali:~/lab#
```

“delete/hide” the node `<Level></Level>`

```
root@kali:~/lab#  
root@kali:~/lab# xmlstarlet ed -N x="http://schemas.microsoft.com/win/2004/08/events/event"  
-d '//x:Level' SecurityEvent1.xml  
<?xml version="1.0"?>  
<Event xmlns="http://schemas.microsoft.com/win/2004/08/events/event">  
  <System>  
    <Provider Name="Microsoft-Windows-Security-Auditing" Guid="{54849625-5478-4994-a5ba-3e3b  
0328c30d}"/>  
    <EventID Qualifiers="">4608</EventID>  
    <Version>0</Version>  
    <Task>12288</Task>  
    <Opcode>0</Opcode>  
    <Keywords>0x8020000000000000</Keywords>  
    <TimeCreated SystemTime="2015-03-25 10:15:35.248869"/>  
    <EventRecordID>1</EventRecordID>  
    <Correlation ActivityID="" RelatedActivityID="">  
    <Execution ProcessID="464" ThreadID="468"/>  
    <Channel>Security</Channel>  
    <Computer>37L4247F27-25</Computer>  
    <Security UserID="">  
  </System>  
  <EventData/>  
</Event>  
root@kali:~/lab#
```

ed: edit
-d: delete

← *Level has been removed*

Omit XML declaration (<?xml ...?>) and “**d**delete” the node <Level></Level>

```
root@kali:~/lab#  
root@kali:~/lab# xmlstarlet ed -0 -N x="http://schemas.microsoft.com/win/2004/08/events/event"  
-d '//x:Level' SecurityEvent1.xml  
<Event xmlns="http://schemas.microsoft.com/win/2004/08/events/event">  
  <System>  
    <Provider Name="Microsoft-Windows-Security-Auditing" Guid="{54849625-5478-4994-a5ba-3e3b0328c30d}" />  
    <EventID Qualifiers="">4608</EventID>  
    <Version>0</Version>  
    <Task>12288</Task>  
    <Opcode>0</Opcode>  
    <Keywords>0x8020000000000000</Keywords>  
    <TimeCreated SystemTime="2015-03-25 10:15:35.248869" />  
    <EventRecordID>1</EventRecordID>  
    <Correlation ActivityID="" RelatedActivityID="" />  
    <Execution ProcessID="464" ThreadID="468" />  
    <Channel>Security</Channel>  
    <Computer>37L4247F27-25</Computer>  
    <Security UserID="" />  
  </System>  
  <EventData />  
</Event>
```

-d: is not **actually** deleting the node
only show deleted stream

How to actually delete a node?

DELETING NAMESPACE
DECLARATIONS

Show the first event and remove error messages

```
root@kali:~/lab#  
root@kali:~/lab# xmlstarlet sel -N x="http://schemas.microsoft.com/win/2004/08/events/event" -t -m  
"//x:Event[1]" -c . SecurityEvt.xml 2>/dev/null > SecurityEvent1.xml  
root@kali:~/lab#  
root@kali:~/lab# cat SecurityEvent1.xml  
<Event xmlns="http://schemas.microsoft.com/win/2004/08/events/event"><System><Provider Name="Micro  
soft-Windows-Security-Auditing" Guid="{54849625-5478-4994-a5ba-3e3b0328c30d}"/>  
<EventID Qualifiers="">4608</EventID>  
<Version>0</Version>  
<Level>0</Level>  
<Task>12288</Task>  
<Opcode>0</Opcode>  
<Keywords>0x8020000000000000</Keywords>  
<TimeCreated SystemTime="2015-03-25 10:15:35.248869"/>  
<EventRecordID>1</EventRecordID>  
<Correlation ActivityID="" RelatedActivityID=""/>  
<Execution ProcessID="464" ThreadID="468"/>  
<Channel>Security</Channel>  
<Computer>37L4247F27-25</Computer>  
<Security UserID=""/>  
</System>  
<EventData/>  
</Event>root@kali:~/lab#
```

Need to remove xmlns because it causes some functions don't work properly

Substitute *xmlns* with nothing (remove *xmlns*)

```
root@kali:~/lab#  
root@kali:~/lab# xmlstarlet ed -0 -N x="http://schemas.microsoft.com/win/2004/08/events/event"  
-d '//x:Level' SecurityEvent1.xml | sed -e 's/xmlns.*=".*"/g'  
<Event > ← xmlns is deleted  
  <System>  
    <Provider Name="Microsoft-Windows-Security-Auditing" Guid="{54849625-5478-4994-a5ba-3e3b0328c30d}"/>  
    <EventID Qualifiers="">4608</EventID>  
    <Version>0</Version>  
    <Task>12288</Task>  
    <Opcode>0</Opcode>  
    <Keywords>0x8020000000000000</Keywords>  
    <TimeCreated SystemTime="2015-03-25 10:15:35.248869"/>  
    <EventRecordID>1</EventRecordID>  
    <Correlation ActivityID="" RelatedActivityID=""/>  
    <Execution ProcessID="464" ThreadID="468"/>  
    <Channel>Security</Channel>  
    <Computer>37L4247F27-25</Computer>  
    <Security UserID=""/>  
  </System>  
  <EventData/>  
</Event>  
root@kali:~/lab#
```

- -e: regular expression
- s/e1/e2/g:
 - s: substitute
 - e1: regex
 - e2: regex
 - g: globe

```
root@kali:~/lab#  
root@kali:~/lab# xmlstarlet ed -O -N x="http://schemas.microsoft.com/win/2004/08/events/event"  
-d '//x:Level' SecurityEvent1.xml | sed -e 's/xmlns.*=".*"//g' > SecurityEvent1_clean.xml  
root@kali:~/lab#  
root@kali:~/lab# cat SecurityEvent1_clean.xml Save the cleaned .xml  
<Event >  
  <System>  
    <Provider Name="Microsoft-Windows-Security-Auditing" Guid="{54849625-5478-4994-a5ba-3e3b0328c30d}"/>  
    <EventID Qualifiers="">4608</EventID>  
    <Version>0</Version>  
    <Task>12288</Task>  
    <Opcode>0</Opcode>  
    <Keywords>0x8020000000000000</Keywords>  
    <TimeCreated SystemTime="2015-03-25 10:15:35.248869"/>  
    <EventRecordID>1</EventRecordID>  
    <Correlation ActivityID="" RelatedActivityID=""/>  
    <Execution ProcessID="464" ThreadID="468"/>  
    <Channel>Security</Channel>  
    <Computer>37L4247F27-25</Computer>  
    <Security UserID=""/>  
  </System>  
  <EventData/>  
</Event>  
root@kali:~/lab#
```

TRANSFORMING AN XML DOCUMENT INTO A CSV

We want to save *EventID* and *ItemCreated* to .csv

```
root@kali:~/lab# /bin/bash 95x28
root@kali:~/lab# cat SecurityEvent1_clean.xml
<Event >
  <System>
    <Provider Name="Microsoft-Windows-Security-Auditing" Guid="{54849625-5478-4994-a5ba-3e3b0328c30d}"/>
    <EventID Qualifiers="">4608</EventID>
    <Version>0</Version>
    <Task>12288</Task>
    <Opcode>0</Opcode>
    <Keywords>0x8020000000000000</Keywords>
    <TimeCreated SystemTime="2015-03-25 10:15:35.248869"/>
    <EventRecordID>1</EventRecordID>
    <Correlation ActivityID="" RelatedActivityID=""/>
    <Execution ProcessID="464" ThreadID="468"/>
    <Channel>Security</Channel>
    <Computer>37L4247F27-25</Computer>
    <Security UserID=""/>
  </System>
  <EventData/>
</Event>
root@kali:~/lab#
```

save to .csv

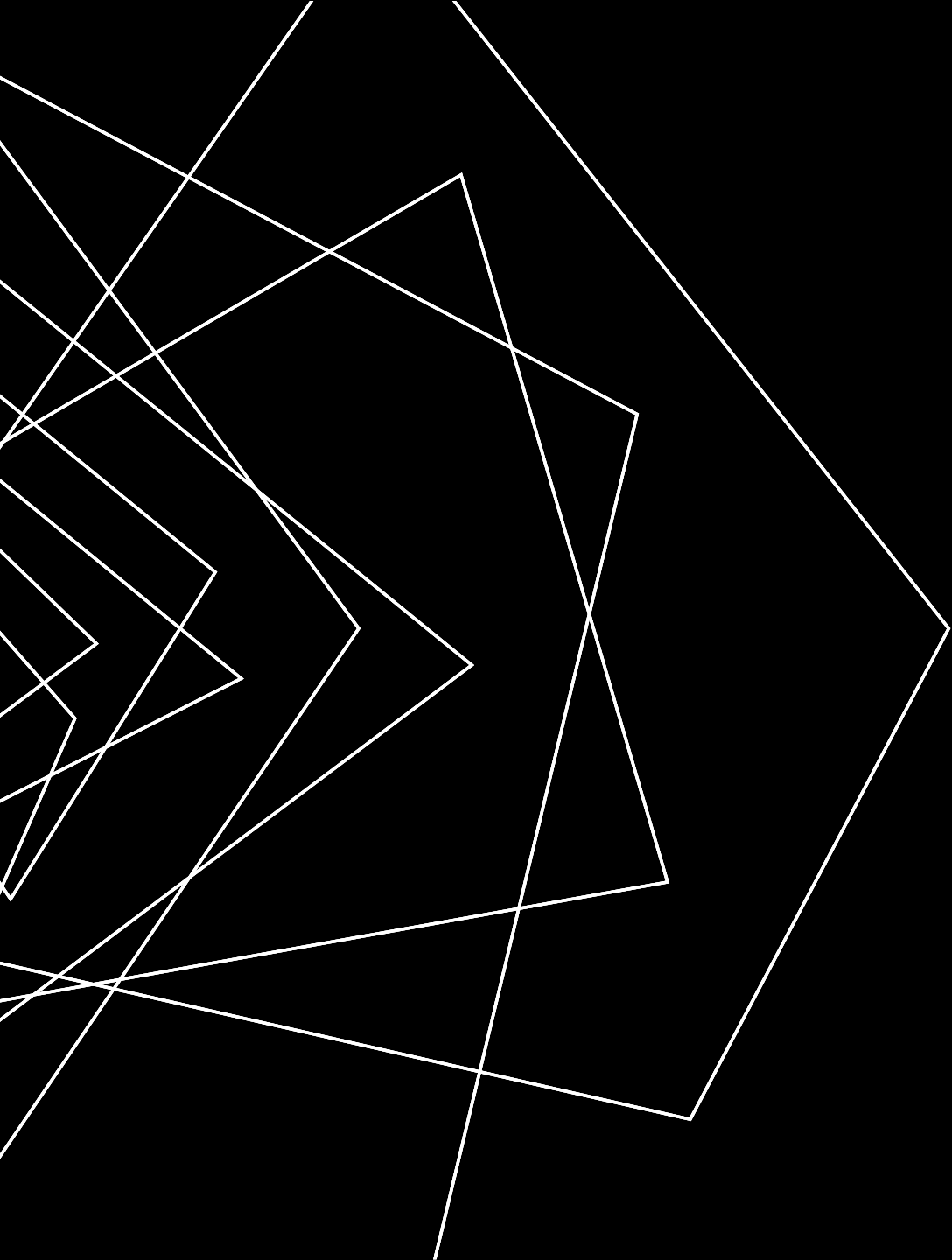
UTC

```
<Event><System><Provider Name="Microsoft-Windows-Security-Auditing" Guid="{54849625-5478-4994-a5ba-3e3b0328c30d}"></Provider>
<EventID Qualifiers="">4608</EventID>
<Version>0</Version>
<Level>0</Level>
<Task>12288</Task>
<Opcode>0</Opcode>
<Keywords>0x8020000000000000</Keywords>
<TimeCreated SystemTime="2015-03-25 10:15:35.248869"></TimeCreated>
<EventRecordID>1</EventRecordID>
<Correlation ActivityID="" RelatedActivityID=""></Correlation>
<Execution ProcessID="464" ThreadID="468"></Execution>
<Channel>Security</Channel>
<Computer>37L4247F27-25</Computer>
<Security UserID=""></Security>
</System>
<EventData></EventData>
</Event>
```

Display *EventID* and *ItemCreated* in text

- T: output is text
- t: template
- m: template
- v: print value of xpath
- n: new line

```
root@kali:~/lab#  
root@kali:~/lab# xmlstarlet sel -T -t -m /Event/System  
  ,TimeCreated/@SystemTime)" -n SecurityEvent1_clean.xml  
4608;2015-03-25 10:15:35.248869  
root@kali:~/lab#
```



THANK YOU

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